Lab Exercise 19- ClusterIP in Service in Kubernetes

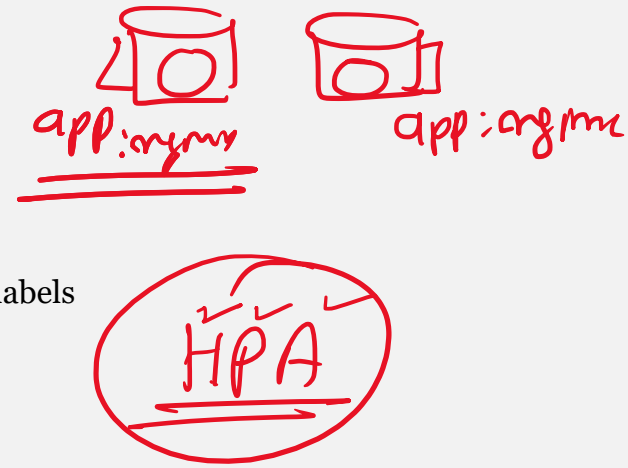
A ClusterIP is the default type of service in Kubernetes. It provides a stable internal IP address for the service, which is accessible only within the cluster. This is useful for internal communication between services within the cluster.

Step 1: Create Deployment (Correct Labels)

Create file:

nginx-deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx   # MUST match template labels
  template:
    metadata:
      labels:
        app: nginx  # MUST match service selector
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```



```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

Save file.

```
kubectl apply -f nginx-deployment.yaml
```

```
C:\Users\shwet>kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-app created
```

Verify:

```
kubectl get pods --show-labels
```

```
C:\Users\shwet>kubectl get pods --show-labels
NAME                          READY   STATUS                      RESTARTS      AGE     LABELS
configmap-pod                 0/1     CreateContainerConfigError  0             3d12h   <none>
my-pod                        1/1     Running                     4 (19m ago)   59d     app=web
nginx-app-86c57bc6b8-bg8xf    1/1     Running                     0             12s     app=nginx,pod-template-hash=86c5
7bc6b8
nginx-app-86c57bc6b8-dq7ps    1/1     Running                     0             12s     app=nginx,pod-template-hash=86c5
7bc6b8
```

You should see:

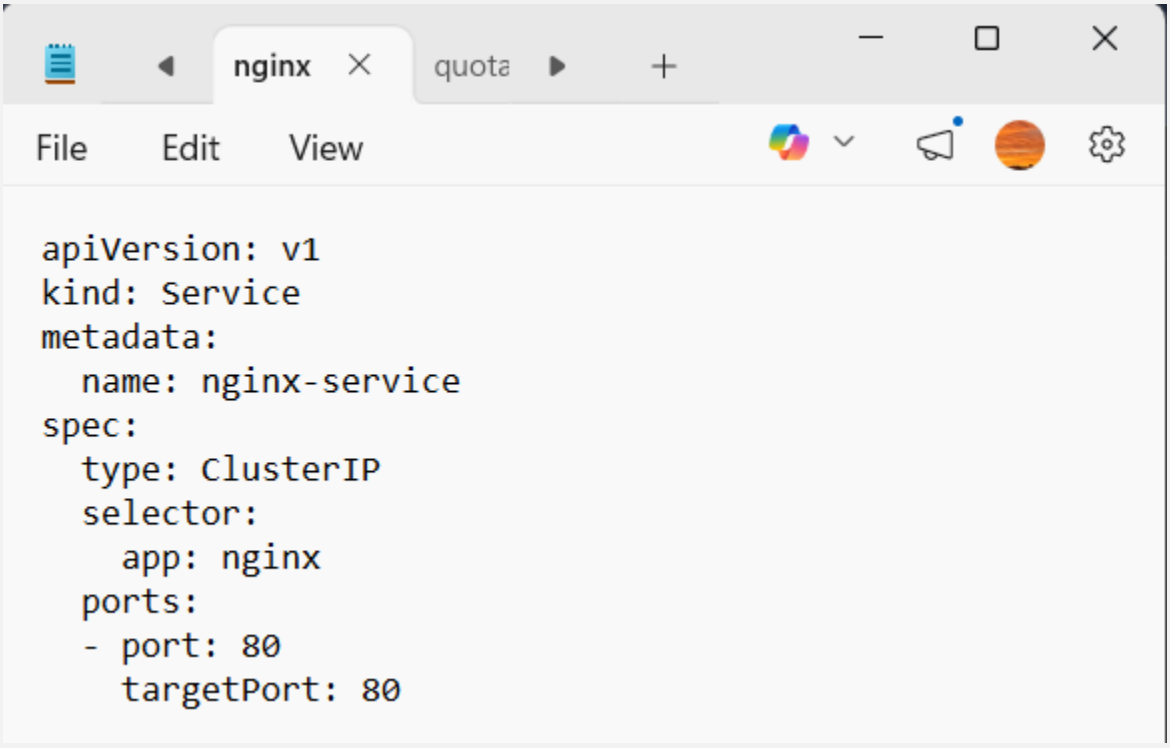
```
app=nginx
```

Step 2: Create ClusterIP Service (Matching Selector)

Create file:

nginx-service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: ClusterIP
  selector:
    app: nginx   # MUST match pod label
  ports:
  - port: 80
    targetPort: 80
```



Apply:

```
kubectl apply -f nginx-service.yaml
```

```
C:\Users\shwet>kubectl apply -f nginx-service.yaml
service/nginx-service created
```

Step 3: Verify It Is Working

Check Service

kubectl get svc

```
C:\Users\shwet>kubectl get svc
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes          ClusterIP   10.96.0.1       <none>        443/TCP        59d
nginx-service       ClusterIP   10.96.254.142   <none>        80/TCP         14s
nodeport-service    NodePort    10.98.155.180   <none>        80:30007/TCP   59d
```

You should see:

TYPE CLUSTER-IP

ClusterIP 10.x.x.x

MOST IMPORTANT CHECK

kubectl get endpoints nginx-service

```
C:\Users\shwet>kubectl get endpoints nginx-service
NAME            ENDPOINTS                       AGE
nginx-service   10.1.0.105:80,10.1.0.106:80   43s
```

Now you should see something like:

10.244.x.x:80, 10.244.x.x:80

Step 4: Test From Inside Cluster

Run test pod:

kubectl run test-pod --image=busybox --rm -it -- sh

Inside pod:

wget -qO- http://nginx-service

```
C:\Users\shwet>kubectl run test-pod --image=busybox --rm -it -- sh
If you don't see a command prompt, try pressing enter.
/ # wget -qO- http://nginx-service
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
```

You should see:

Welcome to nginx!

Exit:

Exit

```
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ # exit
Session ended, resume using 'kubectl attach test-pod -c test-pod -i -t' command when the pod is running
pod "test-pod" deleted
```

Step 5: Test From Outside (It Will Fail)

From your local machine:

```
curl http://<CLUSTER-IP>
```

```
C:\Users\shwet>curl http://<CLUSTER-IP>
The syntax of the command is incorrect.

C:\Users\shwet>
```

It will NOT work.

That proves:

ClusterIP works only inside cluster.